

Knots, Vasocomputation, and the Biology of Stored Tension: A Critical Research Report

Executive summary

The uploaded essay, *Knots of Existence and Vasocomputation*, proposes an unusually ambitious synthesis: myofascial “knots” are interpreted as persistent physiological states; vascular smooth-muscle latch states are proposed as a low-energy storage mechanism; vascular state is then hypothesized to feed back onto neural computation; and Michael Edward Johnson's “vasocomputation” framework is used to connect this loop to predictive processing, trauma, contemplative “knots,” and ultimately consciousness. The essay explicitly acknowledges that this chain begins with established physiology and becomes increasingly speculative; it also states that its text was generated by Kimi and Fable. [filecite turn0file0](#)

The central conclusion of this review is that **several lower-level components of the essay are individually real and scientifically important, but the proposed integration has not yet been demonstrated**. In particular:

1. **Mathematical knot theory is highly developed and genuinely useful in biology**, especially for circular DNA, topoisomerases, viral DNA packing, and knotted proteins. Type-II topoisomerases actively simplify DNA knotting below thermal-equilibrium levels, and confinement inside viral capsids generates measurable knot spectra. ¹ There is, however, **no evidence that a clinical “muscle knot” is a topological knot in the mathematical sense**. The word *knot* currently connects these literatures metaphorically, not mechanistically.
2. **Myofascial trigger points have measurable correlates**, including altered local electrophysiology, chemistry, stiffness, and hemodynamics. Hubbard and Berkoff reported spontaneous needle-EMG activity localized to trigger-point loci; McNulty and colleagues found that mental arithmetic selectively increased activity at trigger points; Shah and colleagues found acidic and mediator-rich biochemical environments; and an imaging systematic review concluded that active and latent trigger points can exhibit quantifiable biomechanical and blood-flow differences. ² However, trigger-point identification by palpation has historically shown limited inter-examiner reliability, and the underlying entity and mechanism remain controversial. ³
3. **The smooth-muscle “latch” phenomenon is established physiology**. Hai and Murphy's kinetic model accounts for force maintenance with reduced cross-bridge cycling after myosin dephosphorylation. But treating this contractile state as a literal long-duration information register—especially one capable of storing semantic predictions, trauma, or “clinging”—is an extrapolation, not a result of the latch literature. ⁴
4. **Bidirectional vascular–neural interaction is more scientifically plausible than it was when Moore and Cao proposed the hemo-neural hypothesis in 2008**. Their hypothesis argued that hemodynamic variables might participate in information processing rather than merely supporting

neuronal metabolism. Subsequent work has shown active neurovascular signaling through capillary endothelium and, particularly strikingly, a 2024 *Science* study demonstrated that arterial pressure pulsations can modulate neuronal activity through mechanosensitive ion channels in rats. ⁵ This supports the general proposition **vascular state → neural effect**. It does not establish the much stronger proposition **vascular latch state = cognitive memory or predictive prior**.

5. The “Michael Johnson” relevant here is **Michael Edward Johnson**, who describes himself as a philosopher and neuroscientist and lists *Vasocomputation* among his major theoretical projects. His 2023–2025 OpenTheory essay proposes the “Compressive Vasomotion,” “Vascular Clamp,” and “Latched Hyperprior” hypotheses, explicitly mapping vascular smooth-muscle contraction onto prediction, medium-term memory, and Buddhist *tanhā/upādāna*. ⁶ Based on the literature located for this review, **Johnson’s vasocomputation work remains a theoretical/self-published research program rather than an experimentally validated, peer-reviewed theory**. Current 2026 Edge City materials describe experiments intended to seek direct evidence, rather than already-established results. ⁷
6. A particularly important neighboring literature actually demonstrates something structurally reminiscent of vasocomputation: **Physarum** slime mold stores information in the geometry of a living flow network. Kramar and Alim experimentally showed that nutrient locations become encoded in a hierarchy of tube diameters, which subsequently biases transport and behavior. ⁸ This is one of the strongest empirical precedents for the general idea that **the state of a biological transport network can encode history and influence subsequent behavior**. It is not evidence that mammalian arteries do the same for thoughts.
7. The uploaded essay’s strongest untested claim is therefore not that blood vessels affect nerves—they do—but that **a spatially localized, history-dependent vascular state can store behaviorally meaningful information and subsequently bias nearby neural computation after the initiating signal disappears**. That claim is experimentally accessible and should be treated as the core vasocomputation hypothesis.
8. The essay’s proposed “breath-release” mechanism is currently far ahead of the data. Slow breathing reproducibly alters cardiovascular autonomic indices such as heart-rate variability, but this does **not** establish that an exhale rapidly unlatches a specific arteriole at an identified trigger point. ⁹ The crucial experiment—simultaneous high-speed measurement of local stiffness, perfusion, respiration, EMG and subjective release—appears not yet to have been performed.
9. A major 2026 development deserves particular attention. Mizrachi, Rottem and Rozenkrantz reported in *Nature Human Behaviour* that voluntarily directing attention to an experimentally inflamed site reduced subsequent histamine-induced inflammatory responses in preregistered within-subject experiments. The study provides unusually strong human evidence for top-down modulation of peripheral physiology, while its authors explicitly note that the molecular/cellular mediators were not directly measured and that extrapolation to chronic disease is uncertain. ¹⁰ This is relevant to vasocomputation because it establishes a controllable cognition-to-periphery channel; it does not establish a vascular latch mechanism.

Overall evidence judgment: the essay is most valuable as a *research-generating synthesis*. Its lower floors—trigger-point physiology, autonomic modulation, smooth-muscle latch kinetics, neurovascular coupling and

biological flow-network adaptation—are grounded in real science. Its middle floor—persistent vascular state influencing neural computation—is plausible enough to test and has increasingly relevant empirical precedents. Its upper floors—muscle knots as latched microvascular memories, thoughts as vascular patterns, contemplative knots as first-person maps of these latches, and enlightenment as systematic unlatching—currently remain speculative.

Evidence scale used in this report

Grade	Interpretation
A — Established/direct	Direct experimental evidence with strong mechanistic grounding and substantial independent support.
B — Supported but bounded	Direct evidence exists, but samples, methods, generality, or replication remain limited.
C — Plausible/indirect	Components are supported, but the specific causal connection has not been demonstrated.
D — Speculative	No direct evidence located for the stated mechanism; proposition is primarily theoretical.

Conceptual, historical, and mathematical foundations

Three meanings of “knot” must be kept separate

The most important conceptual discipline for this subject is to avoid silently moving between three different senses of *knot*.

A **topological knot** is, classically, an embedding of a circle S^1 into three-dimensional space—usually S^3 or $\mathbb{R}^3$ —considered up to ambient isotopy. Two diagrams representing the same tame knot can be transformed into one another through planar deformation plus the three Reidemeister moves. Knot invariants assign quantities that remain unchanged under such equivalences; familiar examples include the fundamental group of the complement, Alexander polynomial and Jones polynomial. ¹¹

A **molecular knot** is a physical polymer configuration that can meaningfully be classified using these ideas. Circular DNA is especially clean mathematically because it is already a closed curve. Protein backbones are open chains, so a closure convention or another operational definition is required; Taylor’s seminal computational study explicitly confronted this issue when identifying deeply knotted proteins. ¹²

A **myofascial “knot”** or trigger point is a clinical/palpatory construct referring to a localized tender or mechanically abnormal region. Nothing in the present trigger-point literature demonstrates that tissue fibers there constitute a nontrivial topological embedding. The uploaded essay’s “knot” is therefore **not currently a knot-theoretic object**. ¹³

This distinction matters because otherwise the rigor of mathematical knot theory can accidentally lend authority to a physiological hypothesis with which it has, so far, no demonstrated topological connection.

The basic mathematics

For a tame knot

$$K : S^1 \hookrightarrow S^3,$$

its complement is

$$X_K = S^3 \setminus K.$$

Much of knot theory asks which properties of X_K , a diagram of K , or algebraic structures derived from it distinguish one isotopy class from another. A knot invariant is a function

$$I : \{\text{knots}\} \rightarrow \mathcal{S}$$

such that

$$K_1 \simeq K_2 \implies I(K_1) = I(K_2).$$

The converse usually fails: most useful invariants are not complete. For example, Alexander polynomials can coincide for distinct knots. ¹⁴

The **crossing number** $c(K)$ is the minimum number of crossings over all diagrams of K . The trefoil has three crossings and the figure-eight knot four. Polynomial invariants encode richer global information than crossing counts. The Alexander polynomial $\Delta_K(t)$, originating in the topology of the infinite cyclic cover of the knot complement, was the first major knot polynomial; Jones introduced a fundamentally new polynomial invariant in the 1980s. ¹⁵

Computationally, knot recognition is nontrivial. Hass, Lagarias and Pippenger showed that the unknot-recognition problem lies in NP and developed exponential worst-case algorithms using normal-surface theory; related topological decision problems have substantial computational complexity. ¹⁶

Computational representations relevant to biology

Model	State representation	Core computation	Typical biological use	Relevance here
Knot diagram / Reidemeister calculus	Crossings with over/under information	Equivalence under local moves	Classification of closed molecular chains	Rigorous for DNA; no demonstrated role in trigger points.
Braid representation	Product of braid generators $\sigma_i^{\pm 1}$	Algebraic manipulation and closure	DNA recombination, molecular topology	Valuable where strands undergo crossing/recombination.

Model	State representation	Core computation	Typical biological use	Relevance here
Polynomial invariants	$\Delta_K(t), V_K(q)$, etc.	Algebraic/ state-sum computation	Identifying knot spectra in polymers	Describes topology, not tissue tone.
Normal-surface algorithms	Triangulated knot complement	Search over normal surfaces	Knot recognition	Computational topology rather than physiological computation. ¹⁶
Polymer Monte Carlo	3-D chain conformations	Sampling knot probabilities under confinement	DNA capsids and polymers	Directly connects mechanics, geometry and knot statistics. ¹⁷
Open-chain knot localization	Protein backbone coordinates plus closure/ localization rule	Identify minimal knotted core	Knotted proteins	Taylor's algorithm established deep protein knots. ¹²
Adaptive-flow-network model	Graph edges with conductance/ tube diameter	Flow-dependent edge reinforcement	<i>Physarum</i> path finding and network memory	Not knot theory, but highly relevant to vasocomputation. ¹⁸
Vasocomputational state model	Neural state + vessel tone/ latch state	Coupled write-store-read dynamics	Proposed mammalian biological computation	Johnson's hypothesis; not yet validated. ¹⁹

The last two rows are particularly important. A **network topology** is not the same mathematical object as a **knot**. A vascular tree can have graph-theoretic topology, cycles and spatial entanglements without containing topological knots. The relevant mathematics for vasocomputation may ultimately be dynamical systems, graph theory, control theory and continuum fluid mechanics more than classical knot invariants.

A useful formalization of the vasocomputation question

Johnson's prose can be translated into a minimal dynamical system without accepting its stronger claims. Let $n_i(t)$ represent neural state near vascular territory i , and $v_i(t)$ an internal vascular state such as smooth-muscle activation, radius or effective resistance. One could write

$$\begin{aligned}\tau_n \dot{n}_i &= F(n_i, u_i, v_i), \\ \tau_v \dot{v}_i &= G(n_i, a(t), m_i) - H(v_i),\end{aligned}$$

where u_i is external neural input, $a(t)$ autonomic/endocrine input, and m_i local metabolic/endothelial signals.

Ordinary neurovascular coupling requires

$$\frac{\partial G}{\partial n_i} / 0. =$$

A hemo-neural effect requires

$$\frac{\partial F}{\partial v_i} / 0. =$$

Vasocomputational memory requires something stronger: v_i must retain information about past inputs,

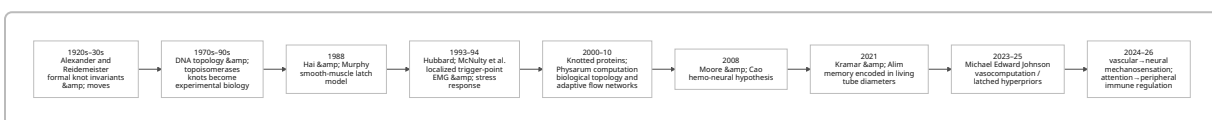
$$v_i(t) / f(u_i(\pm))$$

but instead

$$v_i(t) = \mathcal{F}[\{u_i(s), n_i(s), a(s)\}_{s < t}],$$

and that retained state must later alter neural processing in a reproducible, information-specific fashion. The smooth-muscle latch mechanism could provide hysteresis, but no experiment cited by Johnson or located in this review has yet demonstrated the complete sequence **neural write** → **persistent vascular state** → **later neural readout containing task-relevant information**. The established pieces are therefore necessary but not sufficient. 20

Historical trajectory



The timeline combines several otherwise separate research traditions: mathematical topology, molecular topology, smooth-muscle physiology, trigger-point research, adaptive biological networks, neurovascular neuroscience and Johnson's theoretical synthesis. 21

Biological knots, flow networks, and genuine biological computation

DNA is the strongest bridge between knot theory and biology

DNA topology provides perhaps the clearest example of a topological variable carrying functional biological consequences. Type-II topoisomerases transiently pass one DNA duplex through another, changing knotting and catenation. Rybenkov, Ullsperger, Vologodskii and Cozzarelli found that these enzymes drive knot and catenane fractions as much as approximately 80-fold below equilibrium values, demonstrating active control of a global topological property rather than passive equilibration. 22

This result is interesting computationally because the enzyme does not merely react to a local chemical identity: its activity has consequences for a global property of a polymer. Subsequent theoretical work explored how geometrical selection or kinetic proofreading can permit such topology simplification. ²³

Viral DNA provides another striking experimental case. Arsuaga and colleagues found very high knotting probability in DNA extracted from tailless P4 bacteriophage capsids; Monte Carlo polymer simulations reproduced the tendency of confinement to generate highly knotted molecules. Later analysis of the knot spectrum revealed information about the chiral organization of DNA inside capsids. ²⁴ Here, **knot statistics function as an indirect probe of otherwise inaccessible three-dimensional biological organization**.

This is a legitimate precedent for using topology as an informational observable—but it is categorically different from calling a tender muscular region a knot.

Proteins show that topology can constrain biological dynamics

Taylor's 2000 *Nature* study developed an orientation-independent computational method for locating knots in open protein chains and identified a deep figure-eight knot in acetohydroxy acid isomeroreductase. ¹² Subsequent protein-knot research has examined how topological entanglement affects folding, mechanical stability, enzymatic architecture and degradation. ²⁵

Thus real biological knots can influence kinetics and function. The correct inference is modest but important: **topological state can be biologically consequential**. It does not imply that all persistent biological states called “knots” are topological.

Physarum is the closest empirical analogue to a “vascular memory”

For the vasocomputation question, the slime mold *Physarum polycephalum* may be more relevant than molecular knotting.

Nakagaki, Yamada and Tóth showed that a plasmodial tube network could reorganize to connect food locations by short routes through a maze. ²⁶ Tero, Kobayashi and Nakagaki subsequently modeled its adaptive network with flow-dependent conductance: tubes carrying greater flux thicken and poorly used tubes regress. ²⁷ In simplified form,

$$Q_{ij} = \frac{D_{ij}}{L_{ij}}(p_i - p_j),$$

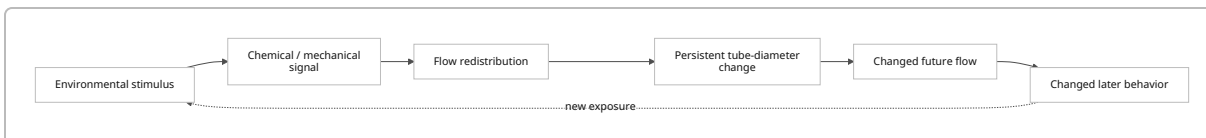
with adaptive conductance

$$\frac{dD_{ij}}{dt} = f(|Q_{ij}|) - \mu D_{ij}.$$

The interaction of conservation laws with positive flow reinforcement performs distributed network optimization without a central controller. Tero and colleagues later showed experimentally and computationally that *Physarum* networks make trade-offs between transport efficiency, construction cost and fault tolerance comparable to engineered transportation networks. ²⁸

Kramar and Alim's 2021 study moves still closer to Johnson's intuition. They observed a local nutrient stimulus changing tube-wall properties and producing network-wide redistribution of tube diameters. The altered hierarchy persisted after the immediate stimulus and biased later migration. Their experimental observations plus model led them to describe the tube-diameter hierarchy as a form of memory. ⁸

The architecture is remarkably suggestive:



This is an **empirically demonstrated write–store–read loop in a living flow network**. ²⁹

It strengthens the *general* scientific plausibility of morphology- or conductance-based biological memory. However, there are decisive differences from mammalian vasculature. *Physarum*'s tubes constitute the organism's own motile transport body, and their remodeling occurs through a specialized cytoskeletal/flow system. Mammalian blood vessels are embedded within a highly differentiated organism whose neural, endothelial, metabolic, endocrine and mechanical controls are much more complex. Therefore, *Physarum* should be treated as a **proof of principle for biological flow-network memory**, not evidence for human vasocomputation.

Neurovascular signaling is genuinely bidirectional

The classical textbook picture in which neural activity controls blood flow is now known to be only part of neurovascular biology. Longden and colleagues demonstrated that brain capillary endothelial cells sense extracellular potassium associated with neural activity and propagate electrical signals upstream to dilate arterioles—a mechanism by which the vascular network actively detects local neural demand. ³⁰

Moore and Cao's 2008 hemo-neural hypothesis went a step further, proposing that blood-flow variables could directly alter neuronal computation through mechanical, thermal, chemical or other pathways. ³¹

A particularly relevant advance came in 2024: Salameh and colleagues reported in *Science* that physiological blood-pressure pulsations influenced neuronal activity in rat brain tissue and awake animals through mechanosensitive ion-channel mechanisms, with effects occurring on rapid timescales. ³² This is genuine evidence for the causal direction

vascular mechanical state → neuronal activity.

The scientific gap is therefore narrower than “blood cannot possibly compute.” The real unresolved question is **what information vascular variables carry, how long it persists, whether it is locally addressable, and whether manipulating that stored vascular state changes cognition in an information-specific way**.

Empirical comparison

System/ study	Data or sample	Experimental method	Principal result	Information- processing relevance	Strength for vasocomputation
Rybenkov et al., 1997	Circular DNA + type-II topoisomerases	Biochemical topology assays	Enzymes reduce knot/catenane abundance far below equilibrium. ²²	Active regulation of global topological state	Indirect precedent
Arsuaga et al., 2002/2005	Bacteriophage DNA	Gel/topological classification + Monte Carlo	Confinement produces high knotting; knot spectrum reveals packing organization. ²⁴	Topology records spatial organization	Indirect precedent
Taylor, 2000	Protein Data Bank structures	Computational knot localization	Deep protein knots objectively identified. ¹²	Topological constraints in biomolecular folding	Indirect
Nakagaki et al., 2000	<i>Physarum</i>	Maze experiments/ video	Living tube network finds efficient connecting paths. ²⁶	Distributed morphological computation	Strong conceptual precedent
Tero et al., 2007/2010	<i>Physarum</i> + model	Flow network modeling and physical networks	Flux-dependent tube adaptation performs transport optimization. ³³	Conductance is both state and computational variable	Strong conceptual precedent
Kramar & Alim, 2021	<i>Physarum</i> time- series imaging	Bright-field imaging + flow/ network model	Tube-diameter hierarchy stores previous nutrient location and biases later behavior. ⁸	Direct morphological memory in a flow network	Closest non- neural analogue
Longden et al., 2017	Mouse brain microvasculature	Electrophysiology/ imaging/ manipulation	Capillary endothelium senses neural- associated K ⁺ and signals upstream. ³⁰	Vascular network actively processes local signals	Supports active vasculature

System/ study	Data or sample	Experimental method	Principal result	Information- processing relevance	Strength for vasocomputation
Salameh et al., 2024	Rat preparations + awake animals	Pressure manipulation/ electrophysiology	Pressure pulsations alter neurons via mechanosensitive channels. ³²	Direct vascular- mechanical → neural signaling	Strong support for “read” path
Johnson, 2023–25	Theoretical synthesis	Conceptual model	VSMC latch proposed as prediction/ hyperprior storage. ¹⁹	Proposed mammalian vascular memory	No direct validation yet

Vasocomputation, Michael Edward Johnson, and adjacent authors

Which Michael Johnson?

The vasocomputation author is **Michael Edward Johnson**. His OpenTheory biography describes his work as spanning philosophy and neuroscience and lists *Principia Qualia* (2016), “Neural Annealing” (2019), “Vasocomputation” (2023), and later AI-consciousness work. He also reports prior involvement with the Qualia Research Institute. ³⁴

Searches also encounter unrelated researchers named Michael or Michel Johnson in neurovascular and biomedical fields. For example, publications by a Michel J. Johnson concern Doppler/vascular-imaging methodology rather than the vasocomputation hypothesis. These should not be conflated with Michael Edward Johnson. The equally important **Michael Levin** discussed in the uploaded essay is another person entirely: Levin is a Tufts developmental biologist studying bioelectric regulation, regeneration and multicellular patterning. ³⁵

Name	Relevant work	Relationship to this report	Priority
Michael Edward Johnson	<i>Principles of Vasocomputation</i> ; OpenTheory	Originator of the specific vasocomputation synthesis discussed by Eugene	Primary ⁶
Michel/Michael J. Johnson and other same-name biomedical authors	Neurovascular/imaging work under similar names	Search disambiguation only; no evidence located that they originated vasocomputation	Low
Michael Levin	Developmental bioelectricity, regenerative pattern memory	Separate researcher; important conceptual neighbor used in Eugene's argument	High, but not “Michael Johnson” ³⁶

Johnson's three central hypotheses

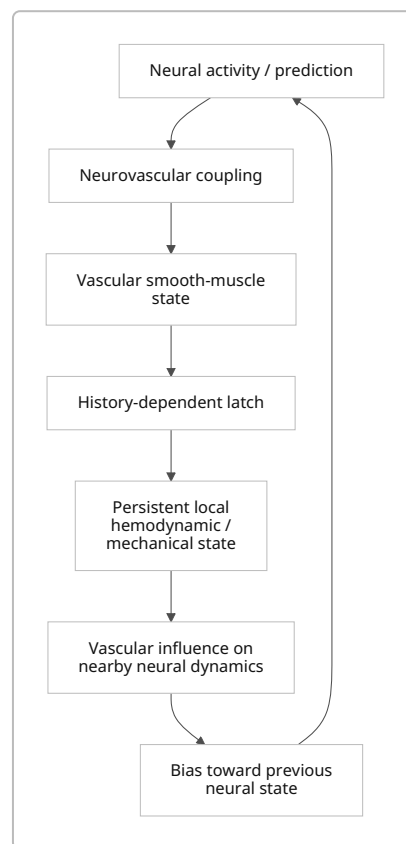
Johnson's *Principles of Vasocomputation* states three especially concrete proposals. ¹⁹

The Compressive Vasomotion Hypothesis proposes that rhythmic vasomotion could operate as a changing mechanical compression field that affects nearby neural resonances.

The Vascular Clamp Hypothesis proposes that vascular contraction can temporarily stabilize or “freeze” nearby neural patterns or plasticity, effectively acting as medium-term memory or a prediction.

The Latched Hyperprior Hypothesis goes further: prolonged vascular smooth-muscle contraction is proposed to engage latch dynamics and thereby maintain nearby circuitry in a constrained state after the original initiating conditions have passed. Johnson maps such persistent constraints onto “hyperpriors” in predictive-processing language and ultimately onto Buddhist concepts of craving or clinging. ¹⁹

The attractive architecture is:



What is experimentally established: neural activity changes vascular state; smooth muscle can maintain force efficiently; vascular variables can feed back onto neurons. ²⁰

What is not established: that the latch stores an identifiable prediction, that it maintains a particular neural representation, or that erasing the vascular state selectively erases or relaxes that representation.

The latch is a mechanism of force maintenance, not yet a memory register

Hai and Murphy's influential model describes smooth-muscle cross-bridge states involving phosphorylation and dephosphorylation. During sustained contraction, force can remain comparatively high while phosphorylation and cross-bridge cycling fall, accounting for economical tonic contraction. ³⁷ Reviews of the latch-bridge hypothesis emphasize both the reality of economical sustained tension and continuing complexity in the precise molecular interpretation. ³⁸

Calling this a **memory** requires an additional operational criterion. In dynamical systems, any hysteretic state can be said to contain information about history. But biological memory relevant to cognition normally requires at least:

state at t depends on prior input,

plus

state can later be read out,

and ideally

different prior inputs produce discriminable later outputs.

The latch physiology directly supports the first criterion in a limited mechanical sense. It does not by itself establish the latter two at a cognitive level.

This is a central place where the uploaded essay's language becomes too strong. Describing vascular latch as a "one-bit storage device distributed through every vascularized tissue" is an illuminating engineering metaphor, but it should not be treated as an experimentally demonstrated storage architecture.

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Other key proponents and adjacent research programs

Christopher Moore and Rosalind Cao are indispensable because their hemo-neural hypothesis provides the most direct conceptual ancestor of Johnson's "read" path. Their 2008 paper explicitly asked whether blood-flow changes may themselves alter information processing. ³¹

Karen Alim and Mirna Kramar provide the strongest experimental example of a living flow-network architecture functioning as memory. Their *Physarum* work should be central to any rigorous future vasocomputation paper, because it supplies a concrete comparator rather than a purely philosophical analogy. ²⁹

Atsushi Tero and Toshiyuki Nakagaki established much of the adaptive-flow-network framework showing how local flow reinforcement can perform globally useful computation without centralized control. ³³

Michael Levin and collaborators demonstrate that non-neural physiological networks can encode persistent pattern information. Durant et al. showed that transient manipulation of endogenous bioelectric gradients in planaria can produce altered regenerative anatomies that persist through subsequent

amputations without continued treatment. ³⁹ This is genuinely fascinating evidence for persistent non-genetic pattern memory, but it is **bioelectric/gap-junction physiology**, not vascular-latch memory.

Levin's group's 2022 adult *Xenopus* study is similarly important but frequently overinterpreted in popular discussions. A 24-hour wearable bioreactor intervention delivering a five-drug regenerative cocktail produced long-lasting tissue repatterning, reinnervation and improved sensorimotor function over many months. ⁴⁰ This demonstrates that a brief intervention can initiate a long biological developmental program. It does **not** demonstrate that cognition, meditation, attention or vascular unlatching can trigger comparable structural regeneration.

Current status of Johnson's program

As of August 2026, the sources located here indicate that Johnson's vasocomputation theory is primarily documented through OpenTheory and related talks/projects rather than a peer-reviewed experimental literature. Current Edge City material describes a "Vasocomputation: The Vascular Imprint of Stress and Trauma" project associated with Michael Johnson and Alex Rockhill, including efforts to determine whether practices such as tension/trauma release alter microvascular tension. ⁷

That distinction is important. There is a legitimate research program here, but the appropriate scholarly wording is presently:

vasocomputation is a testable theoretical synthesis built from established vascular physiology and emerging vascular-neural interaction data, not an experimentally established theory of cognition.

Critical evidence audit of the "Knots of Existence" article

The uploaded article is unusually useful because it explicitly presents itself as a stack of progressively more speculative claims. The main scholarly task is therefore not wholesale acceptance or rejection, but identifying precisely where evidential support changes category. ^{filecite turn0file0}

Myofascial trigger points: real observations, unsettled ontology

Hubbard and Berkoff's 1993 needle-EMG study reported spontaneous activity localized to tiny trigger-point regions but absent in nearby non-tender tissue. ⁴¹ McNulty et al. then studied 14 subjects and found that a mental-arithmetic stressor increased EMG activity specifically at trigger-point sites while nearby control tissue remained comparatively silent. ⁴² This is genuinely intriguing evidence that local trigger-point physiology can be influenced by psychological stress without requiring obvious whole-muscle contraction.

Yet neither experiment identifies the active structure as a blood vessel. The signal was recorded electrophysiologically and has been interpreted in terms of motor endplates, muscle spindles or other neuromuscular sources. Therefore, using these studies as evidence for **"the brain can modulate the state of a trigger point"** is defensible; using them as evidence for **"the brain writes a vascular latch"** is not.

Shah et al. provided a second independent type of evidence. Using microdialysis-like sampling, they reported that active trigger-point sites showed lower pH and increased concentrations of nociceptive/

inflammatory mediators including bradykinin, substance P and CGRP relative to comparison conditions. Their 2008 study involved only nine healthy volunteers, making the result mechanistically interesting but not a definitive population-level characterization. ⁴³

Imaging also supports objective local differences. Mazza et al.'s systematic review included 33 studies and concluded that biomechanical and blood-flow parameters can distinguish active/latent trigger points from healthy tissue, while emphasizing the need for better reproducibility and diagnostic-accuracy research. ⁴⁴

Against this, Lucas et al.'s systematic review found that manual trigger-point examination did not have sufficiently reliable diagnostic performance to recommend it as an objective test, in part because the field lacks a universally accepted reference standard. ⁴⁵ Quintner, Bove and Cohen went further, arguing that the classic trigger-point construct rests on problematic assumptions; Dommerholt and Gerwin subsequently contested that interpretation. ⁴⁶

The uploaded essay therefore overstates one point when it suggests that skeptics do not dispute the existence of precisely localizable palpable spots and disagree only about mechanism. **The reproducibility and ontological status of the spots themselves are part of the debate.**

Fascia and “densification”

Carla Stecco and colleagues have provided substantial anatomical evidence that hyaluronan is associated with fascial interfaces and may contribute to layer gliding; they have proposed that changes in HA aggregation or viscosity could contribute to fascial “densification.” ⁴⁷ This is a plausible mechanism for altered sliding behavior.

However, “fascial densification” should not be treated as an established explanation for trigger points. Some of the evidence is anatomical, cadaveric or mechanistic rather than direct prospective demonstration that a given painful trigger point consists of reversible HA aggregation. Moreover, a quantitative study of HA across fascial sites did not find a simple universal age effect of the kind implied by the essay's broad statement that such densification demonstrably increases with age. ⁴⁸

Does exhalation release knots?

Slow voluntary breathing has reproducible physiological effects. A systematic review by Zaccaro and colleagues found changes in respiratory sinus arrhythmia, heart-rate variability and other autonomic/CNS measures during slow breathing. ⁹

The much more specific claim—

long exhale → parasympathetic shift → local vascular unlatching → instantaneous knot release
—has not been demonstrated.

I did not locate a controlled experiment simultaneously showing a breath-phase-locked step reduction in trigger-point stiffness and increase in microvascular perfusion while pressure, EMG and respiration are independently measured. Thus the essay's repeated observation that “the exhale is the key” should

presently be classified as **phenomenological/practitioner observation requiring objective testing**, not established physiology.

The trigger-point evidence stack

Claim in the essay	Relevant empirical evidence	Main methodological weakness	Assessment
Trigger points show localized spontaneous electrical activity	Needle EMG at trigger-point loci vs adjacent tissue. ⁴¹	Old, small studies; generator not definitively identified	B
Mental stress can increase local trigger-point activity	Mental arithmetic increased TrP EMG in n=14. ⁴²	Small sample; no vascular measurement	B–
Active trigger points have abnormal chemistry	Lower pH and elevated inflammatory/nociceptive analytes. ⁴³	n=9; localization/systemic effects difficult to separate	B–
Trigger points differ in stiffness/flow	Ultrasound/elastography/Doppler literature; systematic review of 33 papers. ⁴⁴	Heterogeneous techniques and definitions	B
Palpation reliably identifies a discrete biological lesion	Reliability literature mixed/weak. ⁴⁵	No gold-standard lesion definition	C–
Fascia HA densification causes the knot	Anatomical and mechanistic HA studies. ⁴⁷	Indirect; trigger-point causal link unproven	C
A knot “releases” objectively in seconds during exhalation	No direct synchronized imaging study located	Core measurement absent	D
The releasing element is vascular smooth muscle	Latch exists; local flow abnormalities exist separately. ⁴⁹	No study identifies VSM latch at a TrP	D

Attention can regulate peripheral biology—but not yet via the proposed latch

The August 2026 *Nature Human Behaviour* study by Mizrachi and colleagues materially changes the landscape because it supplies direct controlled evidence that voluntarily manipulating attention can alter an acute peripheral immune response. In preregistered within-subject experiments, directing attention toward the histamine challenge site produced smaller/faster-resolving reactions than directing attention elsewhere; the authors explored both sensory and autonomic contributions. ¹⁰

This supports:

voluntary cognitive state → measurable localized peripheral physiology.

It does **not** establish:

attention → vascular smooth-muscle latch removal.

The study's authors did not directly measure the relevant molecular immune mediators, nor was the experiment about chronic trigger points or persistent vascular memory. ¹⁰ The uploaded essay is therefore justified in treating the result as an important “existence proof” for top-down peripheral regulation, but goes beyond the data when it folds this immediately into a vasocomputation mechanism.

Where the smooth-muscle argument makes a logical leap

The essay's central mechanistic construction can be decomposed formally:

A : neurons influence vessels

B : smooth muscle exhibits economical persistent force

C : vessels can influence neurons

therefore it proposes

D : vessels store neural information.

Evidence supports *A*, *B*, and increasingly *C*. ⁵⁰ But *D* does not logically follow without showing encoding specificity, persistence, decoding and behavioral consequences.

An ordinary thermostat has hysteresis, but its hysteresis is not therefore autobiographical memory. Likewise, a blood vessel retaining tone after a stimulus contains *some* information about its dynamical history; showing that this constitutes a neurally readable prediction requires additional experiments.

The weakest form of vasocomputation—

vascular variables are part of the state-space that determines neural computation—

is now quite plausible.

The intermediate form—

history-dependent vascular states provide functionally significant slow memory for local neural circuits—

is an excellent experimental hypothesis.

The strongest form—

specific thoughts, clinging, trauma, priors and identity are stored as spatial patterns of vascular latch tension—

has essentially no direct evidence at present.

“Vascular clamp,” hypoxia and confounding

Any experiment purporting to demonstrate vasocomputational memory must distinguish an information-bearing vascular effect from ordinary changes in oxygen, glucose, pH, temperature and waste clearance. Moore and Cao themselves framed multiple possible hemo-neural mechanisms. ³¹

For example, if experimentally constricting a vessel suppresses nearby neural activity, that result alone is not evidence that the vessel is “computing”: reduced perfusion can alter neuronal activity through straightforward metabolic limitation. A convincing vasocomputation experiment must show **structured modulation beyond nonspecific ischemic suppression**, ideally while oxygenation and metabolic variables are separately measured or clamped.

The 2024 demonstration of rapid mechanosensitive effects is consequently important: it identifies one route through which vascular mechanical state can alter neuronal behavior without requiring slow nutrient deprivation. ³²

Contemplative knots and somatic “release”

The uploaded essay draws parallels among Tibetan *rtsa mdud*, Hindu *granthi*, Chinese tender-point traditions, Reichian armor, vipassanā sensations and modern trigger points. ⁵⁰filecite0turn0file0⁵¹ These comparisons may be fruitful as comparative phenomenology, but they are not currently anatomical equivalences.

To elevate them scientifically, one would need blinded historical/anatomical coding: specify traditional locations **before** inspecting modern maps, transform them into a common coordinate system, and test whether they overlap more strongly than chance with vascular branch points, autonomic chains, fascial junctions, trigger-point prevalence or other candidate structures. Without such an analysis, apparent correspondence is vulnerable to post-hoc matching.

The essay is on stronger footing when it says intensive contemplative practice can sometimes produce adverse experiences. Population research has found meditation-related adverse effects ranging from anxiety and traumatic re-experiencing to longer-lasting functional impairment in a minority of practitioners. ⁵¹ That evidence does **not**, however, identify rapid “de-knotting” or vascular unlatching as the mechanism.

Accordingly, the sequence

intensive meditation → adverse autonomic/psychological effects

has empirical support, whereas

adverse effects → mass vascular latch release

is speculative.

The Midjourney ultrasound scanner claim requires correction

The uploaded article says that, by June 2026, a whole-body ultrasonic CT machine effectively exists and suggests that adding Doppler and elastography may be primarily a firmware problem.

Public reporting is substantially more cautious. Midjourney Medical announced a water-immersion full-body ultrasound prototype/development program built around ultrasound-on-chip technology, with very rapid scanning as a target and body-composition imaging among intended applications; the system was not an established clinically validated whole-body perfusion/elastography platform. ⁵²

Therefore:

“There is an announced/prototyped platform that might eventually permit unusually broad ultrasound coverage” is supportable.

“The knot census scanner exists, and Doppler/elastography are merely a firmware update” is not presently established.

In fact, quantitative elastography and microvascular Doppler across a large three-dimensional volume introduce pulse-sequence, frame-rate, calibration, motion, angular-access and validation problems. Whether the same hardware can solve them is an engineering question that should be demonstrated, not presumed.

Michael Levin's regeneration research does not close the cognition-regeneration gap

The essay explicitly labels its regeneration extension speculative, and that caution is justified. Levin's experimental literature provides strong evidence that transient physiological interventions can durably alter regenerative patterning in organisms such as planaria. ⁵³ In adult *Xenopus*, a short, local multidrug treatment produced substantially enhanced long-term regeneration. ⁵⁴

But the bridge

meditative state → vascular unlatching → bioelectric pattern reset → regeneration

contains multiple untested arrows. No controlled evidence cited here demonstrates cognition regenerating lost complex anatomy in humans.

Levin's work is best viewed as showing that **biological tissues possess high-level regulatory dynamics and persistent non-genetic state variables**. That makes broad ideas about somatic information scientifically respectable. It does not validate vasocomputation specifically.

Overall claim matrix

Proposition	Evidence status	Why
Smooth muscle can maintain force with low cycling/energy demand	A	Core smooth-muscle physiology. ⁴
Neural activity regulates local vascular state	A	Neurovascular coupling is extensively established; endothelial sensing mechanisms directly demonstrated. ³⁰
Vascular mechanical state can influence neurons	A/B	Direct mechanosensitive evidence now exists in animal systems. ³²
Biological flow-network morphology can store information	A , in <i>Physarum</i>	Direct experimental memory in tube-diameter hierarchy. ²⁹
Trigger points have objective physiological correlates	B	EMG, chemistry, stiffness and flow findings, but heterogeneous definitions and methods. ⁵⁵
Psychological stress can alter local trigger-point activity	B–	Direct but small, old EMG experiment. ⁴²
Slow breathing alters autonomic physiology	A/B	Substantial breathing/HRV literature. ⁹
An exhale objectively releases an MTrP within seconds	D	Critical time-resolved experiment absent.
MTrPs are latched microvessels	D	No direct visualization/pharmacological identification.
Vascular latch states encode neural predictions	D	Johnson hypothesis; encoding/readout not demonstrated. ¹⁹
Thoughts are spatial patterns of vascular tension	D	No direct neural/vascular decoding evidence.
Traditional contemplative knots correspond anatomically to vascular latches	D	No blinded quantitative mapping.
Attention can modify localized peripheral inflammatory physiology	A/B	Controlled human experiments in 2026. ¹⁰
Meditation-related adverse effects can occur	B	Population and clinical literature supports occurrence. ⁵¹

Proposition	Evidence status	Why
Such adverse effects are caused by mass unlatching	D	No mechanistic evidence.
Bioelectric networks can carry persistent morphological information	A/B , in model organisms	Experimental planarian manipulation. ³⁹
Cognitive states can trigger comparable human structural regeneration	D	Demonstrated ceiling is much lower; no controlled structural-regeneration evidence.

Research program, falsification tests, and open questions

The most productive way to develop vasocomputation is to **remove consciousness, Buddhism and trauma from the first generation of experiments** and first determine whether the proposed physical memory element exists. If the basic mechanism survives, the higher-level claims can then be examined.

Directly image a purported “knot release”

This is the most immediately informative experiment in the uploaded essay, but it needs tighter controls than the essay proposes.

Recruit approximately 40–60 participants who each have at least one reproducibly identifiable symptomatic site and a matched adjacent site. Site localization should be performed independently by two blinded examiners, with pressure algometry used to quantify tenderness and ultrasound/elastography used to predefine the analysis volume rather than letting the operator choose the “best-looking” voxel after the fact. The problems in palpation reliability make this essential. ¹³

Simultaneously record:

Channel	Measurement
Mechanical state	Ultrafast shear-wave elastography
Blood flow	Sensitive power/microvascular Doppler and/or diffuse optical/NIRS measures
Electrical state	Surface plus, in a mechanistic subgroup, fine-wire EMG
Applied force	Instrumented load cell
Respiration	Chest/abdominal inductance + airflow/capnography
Autonomic state	ECG/HRV, skin conductance, beat-to-beat pressure
Subjective event	Button press at perceived “release”
Movement artifact	Probe position/accelerometer tracking

Use a randomized factorial protocol: constant pressure + slow exhalation; constant pressure + normal breathing; slow breathing without local pressure; pressure at matched sham tissue; and unsuccessful release attempts. Slow breathing has known autonomic effects, allowing the key question to be whether the **local** changes exceed global physiological effects. ⁹

The critical preregistered prediction is:

$$t_{\text{release}} \approx t_{\downarrow \text{stiffness}} \approx t_{\uparrow \text{perfusion}},$$

with a consistent temporal ordering across participants.

For the stronger vascular-latch version, the key prediction is

$$t_{\uparrow \text{perfusion}} < t_{\downarrow \text{stiffness}},$$

because vascular relaxation should precede secondary mechanical relaxation.

Falsification criterion: subjective releases occur without reproducible local perfusion or stiffness transitions, or equivalent transitions occur equally at sham sites and during non-release trials.

Test whether vascular state can carry a persistent neural trace

This is the decisive vasocomputation experiment and is more important scientifically than another foam-rolling trial.

In an animal preparation allowing optical access to cortex or olfactory bulb, independently manipulate vascular smooth muscle while recording local neural populations. The 2024 finding that pressure pulsations affect neurons provides a mechanistic starting point. ³²

A modern implementation could combine two-photon vessel-radius measurements, oxygen measurements and neuronal calcium/electrophysiology. Neural pattern A would first be evoked repeatedly while a candidate vascular state is induced. The initiating neural input would then cease. After a delay, test whether the maintained vascular state biases neural responses toward A , compared with identical stimulation under normalized vascular conditions.

Conceptually:

$$A \rightarrow V_A \rightarrow \text{delay} \rightarrow \text{neutral input} \rightarrow A'?$$

The critical controls must maintain comparable oxygenation and rule out nonspecific ischemic suppression. A vascular state that simply reduces all firing is not a memory.

Positive result: different learned vascular states V_A and V_B cause reliably different later neural response distributions even after the initiating neural pattern has disappeared.

Falsification: vascular manipulation affects neural gain only according to instantaneous pressure/perfusion, with no meaningful history dependence once those variables are equalized.

This experiment cleanly distinguishes ordinary neurovascular physiology from genuine vasocomputational memory.

Establish whether vascular smooth-muscle latch is present at trigger points

A trigger point should not be called a vascular latch until the proposed substrate is demonstrated.

The cleanest initial work may be in an animal or ex-vivo preparation where vessel radius, smooth-muscle calcium, myosin-light-chain phosphorylation, perfusion and surrounding tissue mechanics can be measured simultaneously. Candidate trigger-point-like regions could be created through sustained low-level motor activity or experimentally controlled nociceptive/stress paradigms.

Interventions should separately manipulate neural drive and vascular tone. For example, sympathetic blockade, nitric-oxide/cGMP manipulation, or direct smooth-muscle interventions could be used under controlled laboratory conditions to ask whether releasing vascular tone precedes normalization of stiffness and nociceptive signaling. The classic trigger-point literature supplies some rationale for sympathetic involvement, but not proof of a vascular latch. ⁵⁶

Required molecular data would include:

$$[\text{Ca}^{2+}]_{\text{VSMC}}, \quad \text{MLC phosphorylation}, \quad r_{\text{vessel}}, \quad Q_{\text{blood}}, \quad E_{\text{tissue}},$$

where E is an elastographic/mechanical measure.

Expected result under the latch hypothesis: a local region maintains increased vascular resistance despite decay of the initiating activation signal and is selectively normalized by interventions that disrupt smooth-muscle force maintenance.

Falsification: abnormal perfusion is secondary to skeletal-muscle compression, inflammation or other mechanisms and disappears without any evidence of persistent intrinsic VSMC state.

Build a genuine trigger-point natural-history atlas

The uploaded essay is right about one unusually consequential gap: there is no mature, population-scale natural-history map demonstrating how a quantified “knot burden” changes through life.

Existing palpation and imaging studies are not a substitute for this.

A multicenter observational cohort of roughly 500–1,000 participants could stratify by age, sex, activity, pain status and relevant comorbidities. Rather than trying to scan every square centimeter, define a reproducible set of bilateral anatomical sampling sites.

At every location collect pressure-pain thresholds, shear-wave stiffness, tissue thickness/echotexture, Doppler vascular indices and symptom status. A subset could receive MRI/MR elastography for validation. Because existing trigger-point imaging studies show promise but heterogeneity, the imaging algorithm and thresholds should be trained on one cohort and locked before external validation. ⁴⁴

The primary quantities should include not only mean stiffness but spatial heterogeneity:

μ_E , σ_E^2 , spatial autocorrelation, number of focal maxima.

This directly tests the essay's suggestion that "lumpiness" rather than mean tissue stiffness may be the important age-related phenotype.

The study could resolve several currently unsupported statements simultaneously: whether focal abnormalities become more common with age; whether latent and painful sites form separable clusters; whether sites predict later pain; and whether long-term contemplative, athletic or sedentary populations differ.

Compare human vascular memory with established flow-network memory

A sophisticated route to making vasocomputation quantitatively respectable would use *Physarum* as a benchmark rather than merely an analogy.

Kramar and Alim already provide a measurable definition of flow-network memory: prior stimulus alters tube conductance, which persists and biases later flow/behavior. ²⁹ Apply the same information-theoretic questions to vascular data:

$$I(S_{t-\Delta}; V_t)$$

measures how much information present vascular state V_t contains about a past stimulus $S_{t-\Delta}$, while

$$I(V_t; N_{t+\Delta} \mid N_t)$$

asks whether vascular state predicts future neural state beyond what current neural state already predicts.

The strongest criterion for vasocomputational memory would therefore be:

$$I(S_{\text{past}}; V_{\text{present}}) > 0$$

and

$$I(V_{\text{present}}; N_{\text{future}} \mid N_{\text{present}}, S_{\text{present}}) > 0.$$

This provides an experimentally meaningful definition of "write, store, read" that does not depend on metaphor.

Research-priority matrix

Proposed study	Core question	Required data	Positive outcome	What would weaken/falsify the theory?	Priority
Time-locked trigger-point release	Is subjective release accompanied by rapid local vascular/mechanical transition?	Elastography, Doppler/NIRS, EMG, respiration, pressure, HRV	Local perfusion rise and stiffness fall synchronized with release, exceeding controls	No local signature or identical sham effects	Immediate
Vascular → neural memory experiment	Can persistent vessel state retain and later bias a neural representation?	Vessel radius, oxygenation, neural population activity, controlled VSM manipulation	History-specific neural bias after initial input ends	Only instantaneous/metabolic effects	Decisive
Direct latch-mechanism study	Are trigger-point flow abnormalities caused by VSM latch?	VSM Ca ²⁺ /MLC state, perfusion, stiffness, pharmacology	Persistent intrinsic vessel state causally drives local abnormality	Skeletal/inflammatory mechanism explains findings	High
Natural-history atlas	Do objectively defined focal abnormalities accumulate or redistribute with age?	Multi-site elastography, flow, pain threshold, longitudinal phenotype	Reproducible age/site trajectories and phenotypic classes	No stable lesion/phenotype or low reproducibility	High
Information-theoretic flow-memory comparison	Does mammalian vasculature meet the same operational memory criteria as <i>Physarum</i> ?	Past stimuli, vascular state trajectories, future neural activity	Significant history storage and independent predictive readout	Vascular state adds no information beyond current neural/metabolic variables	Foundational

Open questions

The field would advance most rapidly by answering a small number of well-posed questions rather than expanding the phenomenological superstructure.

First, **what is the physical object being palpated at a trigger point?** Existing electrical, biochemical and imaging signals do not yet converge on a single lesion type. ⁵⁷

Second, **does a trigger point possess a reproducible fast state transition?** Without objective time-resolved evidence, “release” remains partly dependent on subjective and practitioner reports.

Third, **can smooth-muscle latch persist on the durations Johnson's theory requires?** Physiological low-energy force maintenance and cognitive memory lasting days, months or years are radically different timescale claims. ⁴

Fourth, **is neural readout of vascular state information-specific?** This is the difference between vascular modulation and vascular computation.

Fifth, **what is actually stored?** Vessel radius, myosin phosphorylation, endothelial state, pericyte state, extracellular matrix mechanics, innervation, receptor expression, local metabolites and structural remodeling can all generate history dependence. Calling all such persistence “the latch” risks collapsing multiple biological timescales into a single mechanism.

Sixth, **what distinguishes an adaptive vascular memory from pathology?** A history-dependent state may be beneficial—local autoregulatory adaptation—or harmful. A viable theory must predict the conditions under which persistence increases versus decreases future performance.

Seventh, **does the traditional contemplative literature contain anatomically falsifiable information?** That question can be studied, but only through preregistered mapping rather than retrospective resemblance.

Epistemic conclusion

The most defensible version of the emerging picture is:

Neural computation is embedded in a larger neurovascular-mechanical system

rather than

Thoughts are already known to be vascular latches.

There is now excellent reason to reject an overly passive conception of vasculature. Vascular endothelial cells sense neural-associated signals; vessels participate actively in neurovascular communication; arterial mechanics can alter neuronal activity; living flow networks in other organisms can store environmental information in their geometry; smooth muscle possesses history-dependent contractile dynamics; and voluntary cognitive states can alter peripheral physiological responses. ⁵⁸

Those findings make **weak vasocomputation**—the proposition that vascular state is a dynamically meaningful component of biological information processing—a serious scientific idea.

They do not yet establish **strong vasocomputation**—the proposition that vascular smooth-muscle latches constitute a distributed medium storing predictions, thoughts, trauma or selfhood.

The striking feature of the hypothesis is that this gap is experimentally bridgeable. Unlike many consciousness theories, vasocomputation predicts measurable intermediate variables: vessel radius, resistance, smooth-muscle biochemical state, tissue stiffness, perfusion, neural responsiveness and hysteresis. A negative result at any major causal link would sharply constrain the theory. A positive demonstration of **write → persistence → content-specific readback** would, conversely, establish a genuinely new biological memory mechanism.

That is the scientifically strongest interpretation of the uploaded essay: not as a completed theory of “knots of existence,” but as a set of unusually concrete hypotheses sitting at the intersection of vascular physiology, mechanobiology, pain science, distributed biological computation and embodied cognition.

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